

Parallel and distributed Computing

Lecture 11

Bazam-e-Jahan

Lecturer

The University of Faisalabad

- Recovery line can be calculated from the checkpoint schedule either using a rollback-dependency graph or a checkpoint graph.
- **Rollback Dependency Graph:**
- Construct the graph
- Perform reachability analysis from the failure states
- The recent states which are unreachable from the failed state form the recovery line.

The rollback propagation Algorithm

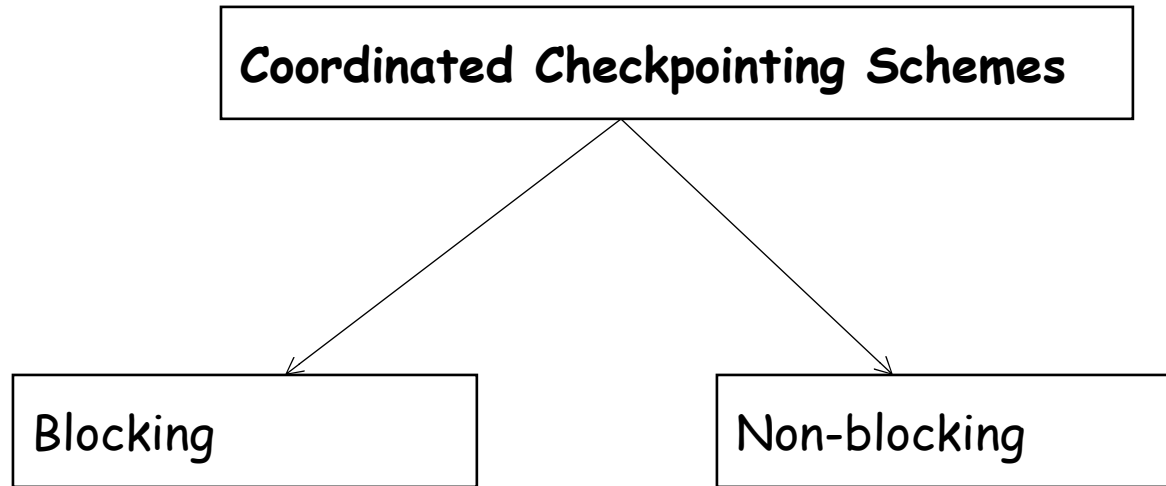
- Step 1: Include last checkpoint of each failed process as an element in set "RootSet";
- Step 2: Include current state of each surviving process as an element in "RootSet";
- While(at least one member of RootSet is marked)
 - Replace each marked element in RootSet by the last unmarked checkpoint of the same process;
 - Mark all checkpoints reachable by following at least one edge from any member of RootSet
- End While
- RootSet is the recovery line

- Any checkpoint that precedes the recovery lines for all possible combinations of process failures can be garbage-collected.
- The garbage collection algorithm based on a rollback dependency graph works as follows:
 - Mark all volatile checkpoints and remove all edges ending in a marked checkpoint, producing a non- volatile rollback dependency graph.
 - Use reachability analysis to determine the worst- case recovery line for this graph, called the global recovery line.



- Coordinated checkpointing requires processes to orchestrate their checkpoints in order to form a consistent global state.
- It simplifies recovery and is not susceptible to the domino effect, since every process always restarts from its most recent checkpoint.
- Only one checkpoint needs to be maintained and hence less storage overhead.
- No need for garbage collection.
- Disadvantage is that a large latency is involved in committing output, since a global checkpoint is needed before output can be committed to the outside world.

Different Coordinated Checkpointing Schemes



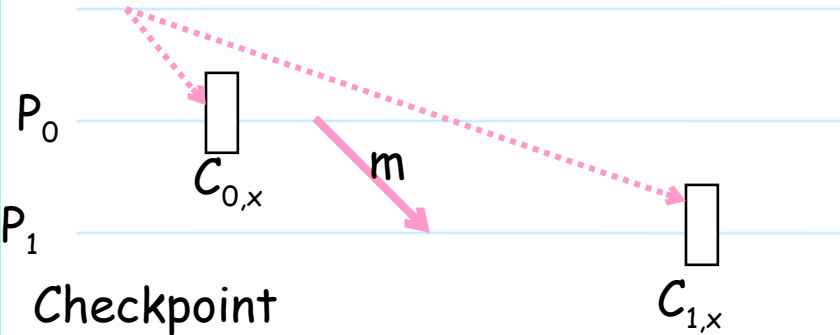
Blocking Coordinated Checkpointing

- **Phase 1:** A coordinator takes a checkpoint and broadcasts a request message to all processes, asking them to take a checkpoint.
- When a process receives this message, it stops its execution and flushes all the communication channels, takes a tentative checkpoint, and sends an acknowledgement back to the coordinator.
- **Phase 2:** After the coordinator receives all the acknowledgements from all processes, it broadcasts a commit message that completes the two-phase checkpointing protocol.
- After receiving the commit message, all the processes remove their old permanent checkpoint and make the tentative checkpoint permanent.
- Disadvantage: Large Overhead due to large block time

- The objective of the coordinated checkpointing is to prevent a process from receiving application messages that could make the checkpoint inconsistent.
- General framework:
 - Checkpoint coordinator / initiator broadcasts the checkpointing message to every other node.
 - Each node upon receiving this message should take a checkpoint.
- However, this approach could lead to checkpoint inconsistency

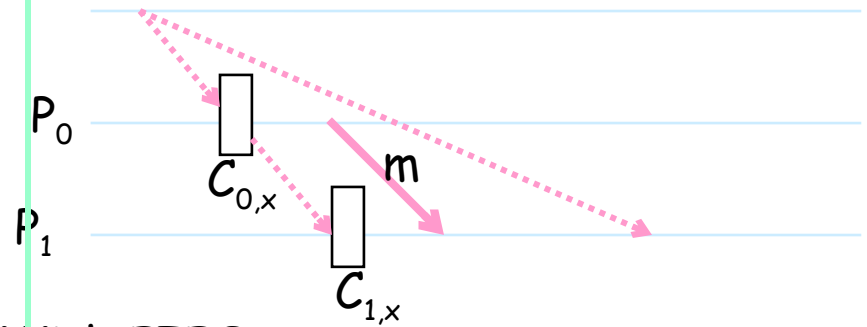
Non-blocking coordinated checkpointing

Initiator



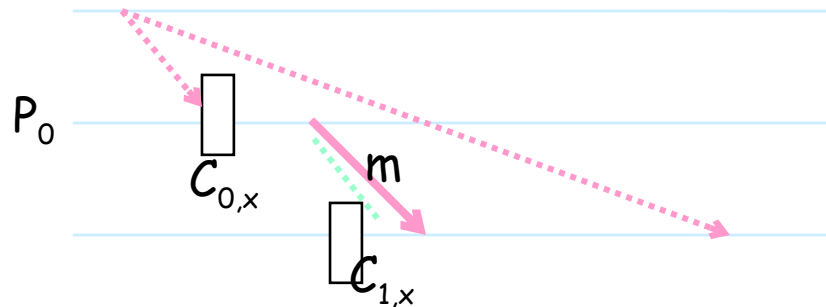
Checkpoint Inconsistency

Initiator



With FIFO channels

Initiator



Non FIFO channels (small dashed line represents the piggybacked checkpoint request)

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.